

GOLDEN RULE: DIMENSIONAL STANDARD != GRADE MARK

DIN 931, DIN 933, ISO 4014, ISO 4017, ASME B18.2.1, IFI-145, etc. define geometry. Strength/material marking comes from ISO 898, ISO 3506, SAE J429/J995, ASTM, EN 14399, etc.

METRIC CARBON / ALLOY STEEL - ISO 898



Common high tensile



Higher strength



Very high strength

Bolt / screw / stud mark: manufacturer identification plus property class, where marking is required and physically possible. Common classes include 4.6, 4.8, 5.8, 8.8, 10.9 and 12.9.

Quick decode (nominal relationship)

8.8 -> nominal tensile 8 x 100 = 800 MPa
Yield ratio 0.8 -> nominal yield about 640 MPa



Full-height nut

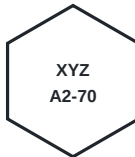


Full-height nut

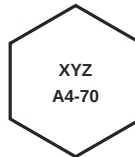


Thin nut class

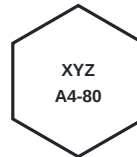
STAINLESS + DIN / EN STRUCTURAL



304-family, common

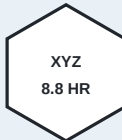


316-family, common



316-family, stronger

ISO 3506 mark = steel grade + property class. A2/A4 indicate stainless grade family; 50/70/80 identify property class. Do not treat "A4" alone as proof of A4-80.



EN 14399-3 System HR

Property class + HR + maker
Do not mix HR and HV components

Legacy callout	Current dimensional reference	Grade marking source
DIN 931 partial thread	DIN EN ISO 4014	ISO 898-1 or ISO 3506-1
DIN 933 full thread	DIN EN ISO 4017	ISO 898-1 or ISO 3506-1
DIN 934 hex nut	DIN EN ISO 4032 / 8673	ISO 898-2 or ISO 3506-2
DIN 6914 structural HV	DIN EN 14399-4	EN 14399 system mark

SAE J429 - EXTERNALLY THREADED FASTENERS



GRADE 2

No radial grade lines

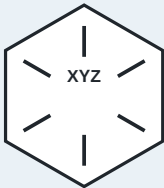
Do not confuse with ungraded / commercial bolt



GRADE 5

3 radial lines

Typical medium-high strength automotive/industrial



GRADE 8

6 radial lines

Higher strength; maker ID also required

SAE J995 - STEEL NUTS



GRADE 5

2 arcs, 120 deg apart



GRADE 8

2 arcs, 60 deg apart

Nut strength belongs to SAE J995, not SAE J429. Manufacturer ID is separate from the grade identifier.

DIRECT APPLICATION - 5/8-11 UNC SERRATED FLANGE NUT

Correct standard stack

Geometry / serration:	IFI-145 or customer drawing. ASME B18.16.4 only for its specified 90,000 psi locknut product.
Thread:	5/8-11 UNC-2B
Mechanical grade:	SAE J995 Grade 8
Mark:	Maker ID + Grade 8 arcs (60 deg apart)
Finish / packing:	Plain/oiled; max 16 kg per carton - state separately

IDENTICAL-LOOKING MARKS CAN MEAN DIFFERENT STANDARDS

Visual mark	May indicate	Why PO / certificate matters
3 radial lines	SAE J429 Grade 5; ASTM A449 Type 1 in applicable size range	Same visual family, different standard scope and size range
6 radial lines	SAE J429 Grade 8; ASTM A354 Grade BD in applicable size range	Do not identify solely from radial lines
No radial lines	SAE Grade 2, ungraded/commercial, ASTM A307 Grade A or exempt product	Maker mark, packaging and MTR are decisive
A325 / A490 text	ASTM F3125 structural bolt grade	Type 3 weathering grade uses an underlined grade symbol

ASTM F3125 STRUCTURAL BOLTS + ASTM A563 NUTS



Type 1



Type 3



Type 1



Type 3



A563 Grade DH



Twist-off assembly



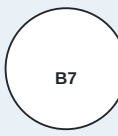
Twist-off assembly



Type 3 nut

ASTM A193 / A194 PRESSURE BOLTING

External bolting - head or stud end



Nuts - grade symbol is part of identity



M is part of the ASTM grade symbol - not a metric-thread mark. Additional class/condition marks may apply.

OTHER COMMON ASTM MARKS

Standard / product	Typical grade mark	Procurement caution
ASTM F1554 anchor rod	36 / 55 / 105 when permanent marking is specified	Permanent manufacturer/grade ID is a supplementary requirement - call it up explicitly
ASTM F593 stainless bolt	F593C, F593D (304-family); F593G, F593H (316-family)	Letter encodes alloy group, condition and diameter range - not just chemistry
ASTM F594 stainless nut	F594C / D; F594G / H, as applicable	Match alloy group and proof strength to the bolt
ASTM A307 general bolt	Grade B commonly marked B; Grade A may be unmarked/exempt	Do not identify Grade A merely from absence of radial lines

STRUCTURAL ASSEMBLY RULES - AUSTRALIA / EUROPE

AS/NZS 1252.1	Australian structural bolt assembly. Property class 8.8 is based on AS 4291.1 (ISO 898-1 equivalent); nut class 8 is based on AS 4291.2.
Accepted alternative	EN 14399-3 System HR class 8.8 is the nominated alternative assembly; HR 10.9 is an additional assembly type when specified.
Do not mix systems	AS/NZS 1252.1 components and EN 14399 HR components are functional assemblies and must not be interchanged within one assembly.
HV warning	System HV is a different ductility system and is not the nominated AS/NZS 1252.1 alternative.
Receiving	Check maker mark, class/system mark, package lot, SDoC/DoP, EN 10204 3.1 report and verification requirement.

5-STEP RECEIVING CHECK

1

Read the PO

Product geometry, thread, grade, coating, edition and certification requirements.

2

Identify all marks

Manufacturer, grade/property class, structural system, type/class and special symbols.

3

Match the package

Lot/heat/batch identification must link product to packing label and certificate.

4

Review certificates

MTR / EN 10204 3.1 / SDoC / DoP / test report must show required tests and results.

5

Escalate ambiguity

Quarantine mixed, illegible, mismarked or undocumented fasteners; do not infer grade by appearance.

WHAT THE MARK DOES NOT TELL YOU

Coating / finish

Zinc, HDG, mechanical zinc, cadmium, PTFE, plain/oiled and passivation are usually not reliably identified from the grade mark.

Thread details

UNC vs UNF, coarse vs fine, tolerance class and left-hand thread require physical inspection and documents.

Exact material heat

A2/A4 or B7/B8M identifies a grade family/class, not the heat chemistry or traceability record.

Full compliance

Correct-looking stamps can be counterfeit or applied to nonconforming product. Tests and traceable records remain necessary.

Application suitability

A grade mark does not establish corrosion, temperature, fatigue, hydrogen embrittlement or joint-design suitability.

PRIMARY / OFFICIAL REFERENCES USED

- ISO 898-1:2013 + Cor 1 - carbon/alloy steel bolts, screws and studs (revision under development).
- ISO 898-2:2022 - carbon/alloy steel nuts.
- ISO 3506-1:2020 and ISO 3506-2:2020 - stainless bolts/studs and nuts.
- DIN Media status pages - DIN 931 / 933 / 934 / 6914 withdrawn and replacement standards.
- EN 14399-3 and EN 14399-4 - preloaded structural assemblies, systems HR and HV.
- AS/NZS 1252.1:2016 and Australian Steel Institute TN001 Version 5.
- SAE J429 MAY2014 - externally threaded inch-series steel fasteners.
- SAE J995 JUL2017 - steel nuts.
- ASTM F3125/F3125M-25a; ASTM A563; RCSC 2020 specification and required-mark figure.
- ASTM A193/A193M-26; ASTM A194/A194M-26; ASTM F593-24; ASTM F594-24; ASTM F1554-20.
- U.S. Customs and Border Protection, Fasteners of Heading 7318 (2012), representative grade-mark chart.
- IFI-145 for serrated flange nut geometry. ASME B18.16.4 is a 90,000 psi locknut product standard and should not be treated as a generic SAE Grade 8 geometry reference.

Official source domains: [iso.org](https://www.iso.org) | [dinmedia.de](https://www.dinmedia.de) | [steel.org.au](https://www.steel.org.au) | [sae.org](https://www.sae.org) / [saemobilus.sae.org](https://www.saemobilus.sae.org) | [store.astm.org](https://www.store.astm.org) | [boltcouncil.org](https://www.boltcouncil.org) | [govinfo.gov](https://www.govinfo.gov) | [asme.org](https://www.asme.org)